



VRIJE
UNIVERSITEIT
BRUSSEL



IMPLEMENTING A BENCHMARK IN BENCHKIT

Performance-Analysis-and-Evaluation

G rard Lichtert
0557513
gerard.lichtert@vub.be

May 10, 2026

Sciences and Bio-Engineering Sciences

Contents

1	Introduction	2
2	Parameters	2

1 Introduction

`memtier_benchmark` is a high-performance traffic generation tool specifically designed to stress-test NoSQL key-value stores like Redis, Valkey, and Memcached. While many benchmarks focus on how fast a database can write to a disk, `memtier` focuses on high-throughput network I/O and concurrency. It simulates thousands of simultaneous clients to measure how well a database handles a massive volume of requests and at what point the response time (latency) begins to degrade.

2 Parameters

The parameters that can be passed along are: